

Wireshark Practical for Windows

Practical 1: Installation of Wireshark

Objective

To install and configure Wireshark on Windows.

Steps

1. Download Wireshark from [Wireshark Official Website](https://www.wireshark.org)
2. Run the installer.
3. Install **Npcap** when prompted.
4. Complete the installation.
5. Launch Wireshark.

Result

Wireshark was successfully installed and ready for packet capturing.

Practical 2: Capturing Network Traffic

Objective

To capture packets on a network interface.

Steps

1. Open Wireshark.
2. Select the active network adapter (Wi-Fi/Ethernet).
3. Click the blue shark-fin icon (Start Capture).
4. Open a website in a browser.
5. Observe packets appearing in real time.
6. Click the red square icon to stop capturing.

Result

Network packets were successfully captured and displayed.

Practical 3: Filtering Packets

Objective

To analyze specific network traffic using filters.

Common Filters

Filter	Description
ip	Show IP packets
tcp	Show TCP packets
udp	Show UDP packets
dns	Show DNS traffic
http	Show HTTP traffic
icmp	Show Ping packets
tcp.port == 80	Traffic on port 80
ip.addr == 192.168.1.1	Specific IP address

Steps

1. Start packet capture.
2. Enter a filter in the filter bar.
3. Press Enter.
4. Observe filtered packets.

Result

Only desired packets were displayed.

Practical 4: Ping Analysis

Objective

To capture and analyze ICMP packets.

Steps

1. Start Wireshark capture.
2. Open Command Prompt.
3. Execute:

ping google.com

4. In Wireshark apply:

icmp

5. Observe Echo Request and Echo Reply packets.

Result

ICMP communication between the host and destination was analyzed.

Practical 5: DNS Analysis

Objective

To observe DNS name resolution.

Steps

1. Start packet capture.
2. Open a website not recently visited.
3. Apply filter:

dns

4. Locate DNS Query and DNS Response packets.

Information Obtained

- Domain Name
- DNS Server IP
- Response Time
- Resolved IP Address

Result

DNS resolution process was successfully observed.